

#14Days14Chapters

DAY 14

[Aim: 100/100 in Maths]

ARITHMETIC PROGRESSION

Arithmetic Progression (A.P.) :-

↳ A sequence in which each term is obtained by adding a fixed number to the previous term

eg:- AP: a_1 3, a_2 5, a_3 7, a_4 9, a_5 11, a_6 13, ...

Arrows between terms are labeled with $+2$.

common → fixed no. = 2

difference (d)

common (d) = $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = a_5 - a_4$

diff

→ $[3-1, 5-3, 7-5, 9-7]$ → all c.d. will be equal in AP.

✓ General form of A.P. → $a, a+d, a+2d, a+3d, a+4d, \dots$

to find a given sequence is A.P. or not??

આપણે જાણીએ કે $a_n - a_{n-1}$ equal હો



is.

all 'd' must be equal.

#LP:- $\textcircled{4}, 8, 12, 16, 20, 24, \dots$, A.P or not?
 $a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6$

$$\begin{aligned} d_1 &= a_2 - a_1 \Rightarrow 8 - 4 \Rightarrow \textcircled{4} \\ d_2 &= a_3 - a_2 \Rightarrow 12 - 8 \Rightarrow \textcircled{4} \\ d_3 &= a_4 - a_3 \Rightarrow 16 - 12 \Rightarrow \textcircled{4} \\ d_4 &= a_5 - a_4 \Rightarrow 20 - 16 \Rightarrow \textcircled{4} \\ d_5 &= a_6 - a_5 \Rightarrow 24 - 20 \Rightarrow \textcircled{4} \end{aligned}$$

$$d_1 = d_2 = d_3 = d_4 = d_5 = \dots$$

all c.d. are equal.

$\therefore$ Yes, it's an A.P.

$$\begin{aligned} a &= 4 \\ d &= 4 \end{aligned}$$

#LP: Which of the following sequences are in arithmetic progression (AP)? For those that are in AP? Find the first term and the common difference.

(i) 3, 6, 12, 24, ...

a_1 a_2 a_3 a_4

$$\left. \begin{aligned} a_2 - a_1 &= 6 - 3 \Rightarrow \textcircled{3} \\ a_3 - a_2 &= 12 - 6 \Rightarrow \textcircled{6} \end{aligned} \right\} \neq \text{Not an A.P.}$$

$$a_4 - a_3 =$$

(ii) 0, -4, -8, -12, ...

a_1 a_2 a_3 a_4

$$a_2 - a_1 = (-4) - 0 \Rightarrow \textcircled{-4}$$

$$a_3 - a_2 = (-8) - (-4) \Rightarrow -8 + 4 \Rightarrow \textcircled{-4}$$

$$a_4 - a_3 = (-12) - (-8) \Rightarrow -12 + 8 \Rightarrow \textcircled{-4}$$

} all c.d are equal.
Yes, it's an AP

$$\begin{aligned} a &= 0 \\ d &= -4 \end{aligned}$$

(iii) $1/2, 1/4, 1/6, 1/8, \dots$

a_1 a_2 a_3 a_4

$$a_2 - a_1 \Rightarrow \frac{1}{4} - \frac{1}{2} \Rightarrow \frac{1-2}{4} \Rightarrow \textcircled{-\frac{1}{4}}$$

$$a_3 - a_2 \Rightarrow \frac{1}{6} - \frac{1}{4} \Rightarrow \frac{4-6}{2 \cdot 4} \Rightarrow -\frac{2}{8} \Rightarrow \textcircled{-\frac{1}{4}}$$

Not an AP.

#LP: $N: 1, 2, 3, 4, 5, 6, 7, 8 \dots$ AP?? find d ??
 $a_1 \ a_2 \ a_3 \ a_4$

$$\left. \begin{array}{l} a_2 - a_1 \Rightarrow 2 - 1 \Rightarrow 1 \\ a_3 - a_2 \Rightarrow 3 - 2 \Rightarrow 1 \\ a_4 - a_3 \Rightarrow 4 - 3 \Rightarrow 1 \end{array} \right\} d = \text{same. } \textcircled{\text{Yes}} \quad \textcircled{d=1}$$

#LP: $2, 2, 2, 2, 2, 2, 2 \dots$ AP? d ??
 $a_1 \ a_2 \ a_3 \ a_4$

$$\left. \begin{array}{l} a_2 - a_1 \Rightarrow 2 - 2 \Rightarrow 0 \\ a_3 - a_2 \Rightarrow 2 - 2 \Rightarrow 0 \\ a_4 - a_3 \Rightarrow 2 - 2 \Rightarrow 0 \end{array} \right\} \text{Same} \\ \text{Yes AP! !}$$

$$\boxed{d=0}$$

#LP:

If $-\frac{5}{7}$, a , 2 are consecutive terms in an Arithmetic Progression, then the value of ' a ' is

(a) $\frac{9}{7}$

(b) $\frac{9}{14}$

(c) $\frac{19}{7}$

(d) $\frac{19}{14}$

$-\frac{5}{7}, a, 2 \rightarrow \text{A.P.}$
 a_1, a_2, a_3

given $\rightarrow$ AP $\therefore$ all c.d are equal

$$a_2 - a_1 = a_3 - a_2$$

$$a - \left(-\frac{5}{7}\right) = 2 - a$$

$$a + \frac{5}{7} = 2 - a$$

$$a + a = 2 - \frac{5}{7}$$

$$2a = \frac{14 - 5}{7}$$

$$2a = \frac{9}{7}$$

$$a = \frac{9}{14}$$

#LP:

Show that $(a - b)^2$, $(a^2 + b^2)$ and $(a + b)^2$ are in A.P. (2020)

$$\begin{aligned}d_1 &= a_2 - a_1 \\&= (a^2 + b^2) - (a - b)^2 \\&= a^2 + b^2 - [a^2 + b^2 - 2ab] \\&= \cancel{a^2} + \cancel{b^2} - \cancel{a^2} - \cancel{b^2} + 2ab\end{aligned}$$

$$\boxed{d_1 = 2ab}$$

$$\begin{aligned}d_2 &= a_3 - a_2 \\&= (a + b)^2 - (a^2 + b^2) \\&= (a^2 + b^2 + 2ab) - a^2 - b^2 \\&= \cancel{a^2} + \cancel{b^2} + 2ab - \cancel{a^2} - \cancel{b^2}\end{aligned}$$

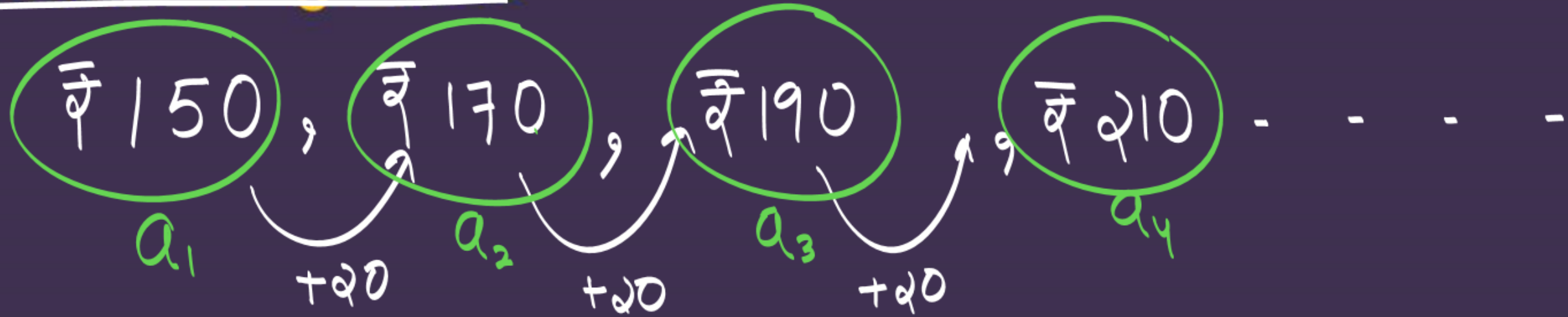
$$\boxed{d_2 = 2ab}$$

Since $d_1 = d_2$ all c.d. are equal

Yes, A.P. ✓

#LP: In which of the following situations the sequence of numbers formed will form an A.P.?

(i) The cost of digging a well for the first metre is Rs 150 and rises by Rs 20 for each succeeding metre.



$$a_2 - a_1 = 170 - 150 = 20$$

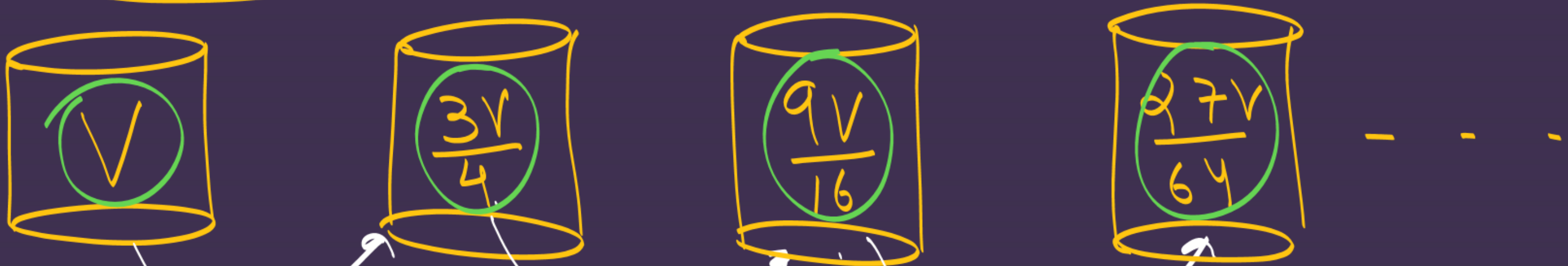
$$a_3 - a_2 = 190 - 170 = 20$$

$$a_4 - a_3 = 210 - 190 = 20$$

all r.d are equal.

Yes, AP ✓✓.

(ii) The amount of air present in the cylinder when a vacuum pump removes each time $\frac{1}{4}$ of the remaining in the cylinder.



$$\begin{aligned}
 &V - \frac{1}{4}V \\
 \rightarrow &\frac{4V - V}{4} \\
 &\text{• } \left(\frac{3V}{4}\right)
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow \frac{3V}{4} - \frac{1}{4}\left(\frac{3V}{4}\right) \\
 &\rightarrow \frac{3V}{4}\left(1 - \frac{1}{4}\right) \\
 &\rightarrow \frac{3V}{4}\left(\frac{3}{4}\right) \rightarrow \frac{9V}{16}
 \end{aligned}$$

$$\begin{aligned}
 &\frac{9V}{16} - \frac{1}{4}\left(\frac{9V}{16}\right) \\
 &\rightarrow \frac{9V}{16}\left(1 - \frac{1}{4}\right) \\
 &\rightarrow \frac{9V}{16} \times \frac{3}{4} \rightarrow \frac{27V}{64}
 \end{aligned}$$

$$V, \frac{3V}{4}, \frac{9V}{16}, \frac{27V}{64} \dots \text{AP?}$$

a_1 a_2 a_3 a_4

$$a_2 - a_1 = \frac{3V}{4} - V \Rightarrow \frac{3V - 4V}{4} \Rightarrow \left(-\frac{V}{4} \right)$$

$$a_3 - a_2 = \frac{9V}{16} - \frac{3V}{4} \Rightarrow \frac{9V - 12V}{16} \Rightarrow \left(-\frac{3V}{16} \right)$$

Not equal.

Not an AP.

(iii) Divya deposited Rs 1000 at compound interest at the rate of 10% per annum.
The amount at the end of the first year, second year, third year, ..., and so on.

$$P = ₹ 1000/-$$

$$R = 10\%$$

$$\text{1st year} \rightarrow A_1 = 1000 \left(1 + \frac{10}{100}\right)^1$$

$$\text{2nd year} \rightarrow A_2 = 1000 \left(1 + \frac{10}{100}\right)^2$$

$$\text{3rd year} \rightarrow A_3 = 1000 \left(1 + \frac{10}{100}\right)^3$$

$A_1, A_2, A_3, \dots$ AP?

$$d_1 = A_2 - A_1$$

$$= 1000 \left(1 + \frac{10}{100}\right)^2 - 1000 \left(1 + \frac{10}{100}\right)^1$$

$$= 1000 \left(1 + \frac{10}{100}\right) \left[\left(1 + \frac{10}{100}\right) - 1\right]$$

$$d_1 = 1000 \left(1 + \frac{10}{100}\right) \left(\frac{10}{100}\right)$$

$$d_2 = A_3 - A_2$$

$$= 1000 \left(1 + \frac{10}{100}\right)^3 - 1000 \left(1 + \frac{10}{100}\right)^2$$

$$= 1000 \left(1 + \frac{10}{100}\right)^2 \left[1 + \frac{10}{100} - 1\right]$$

CI

$$A = P \left(1 + \frac{R}{100}\right)^T$$

$$d_2 = 1000 \left(1 + \frac{10}{100}\right)^2 \left(\frac{10}{100}\right)$$

$$d_1 \neq d_2$$

Not an AP!

$$a_1 = a = a + (1-1)d$$

$$a_2 = a + d = a + (2-1)d$$

$$a_3 = a + 2d = a + (3-1)d$$

$$a_4 = a + 3d = a + (4-1)d$$

$$a_5 = a + 4d = a + (5-1)d$$

$$\vdots$$
$$a_{10} = a + (10-1)d$$

$$a_{1000} = a + (1000-1)d$$

$$\boxed{a_n = a + (n-1)d}$$

n^{th} term of AP

nth term of AP:

$$a_n = a + (n-1)d$$

first term

c.d.

LP:- For an AP, first term is 4 & common diff is -1.
Find the 101th term of This AP

$$a_{101} = a + (101-1)d$$

$$a_{101} = 4 + 100(-1)$$

$$a_{101} = 4 - 100$$

$$a_{101} = -96$$

#LP: If the 17th term of an A.P. exceeds its 10th term by 7, find the common difference. (AI 2019)

Ans.

$$Q_{17} = Q_{10} + 7$$

$$a + (17-1)d = a + (10-1)d + 7$$

$$16d = 9d + 7$$

$$16d - 9d = 7$$

$$7d = 7$$

$$\boxed{d=1}$$

$$d = ??$$

#LP:

Write the nth term of the A.P. $\frac{1}{m}, \frac{1+m}{m}, \frac{1+2m}{m}, \dots$ (CBSE 2017)

$$a_n = a + (n-1)d$$

$$a_n = \frac{1}{m} + (n-1)(1)$$

$$a_n = \frac{1}{m} + n - 1$$

$$a_n = \frac{1 + nm - m}{m}$$

$$d = \left(\frac{1+m}{m} \right) - \left(\frac{1}{m} \right)$$

$$d = \frac{\cancel{1} + m - \cancel{1}}{m}$$

$$d = \frac{\cancel{m}}{\cancel{m}} = 1$$

#LP:

The next term of the A.P. $\sqrt{7}, \sqrt{28}, \sqrt{63}, \dots$

[CBSE 2014]

$$\sqrt{7}, \sqrt{28}, \sqrt{63} \dots$$

$$d = 2\sqrt{7} - \sqrt{7}$$

$$d = \sqrt{7}$$

AP: $\sqrt{7}, 2\sqrt{7}, 3\sqrt{7}, \bigcirc$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $a_1 \quad a_2 \quad a_3 \quad a_4$

$$a_4 = a + (4-1)d$$

$$a_4 = \sqrt{7} + 3(\sqrt{7})$$

$$a_4 = 4\sqrt{7}$$

$$= \sqrt{4^2 \cdot 7}$$

$$= \sqrt{16 \times 7}$$

$$\sqrt{112} = a_4$$

$$28 = 2 \cdot 2 \cdot 7$$

$$\sqrt{28} = \sqrt{2^2 \cdot 7}$$

$$\sqrt{28} = 2\sqrt{7}$$

$$\sqrt{63} = \sqrt{3^2 \cdot 7}$$

$$\sqrt{63} = 3\sqrt{7}$$

$$\begin{array}{r|l} 2 & 28 \\ \hline 2 & 14 \\ \hline 7 & 7 \\ \hline \end{array}$$

$$\begin{array}{r|l} 3 & 63 \\ \hline 3 & 21 \\ \hline 7 & 7 \\ \hline \end{array}$$

#LP: If the 9th term of an A.P. is zero, then show that its 29th term is double of its 19th term. (2019C)

given $\rightarrow$

$$a_9 = 0$$

$$a + (9-1)d = 0$$

$$a + 8d = 0$$

$$a = -8d$$

RHS

$$2a_{19}$$

$$\Rightarrow 2(a + 18d)$$

$$\Rightarrow 2(-8d + 18d)$$

$$\Rightarrow 2(10d)$$

$$\text{RHS} = 20d$$

To prove

$$a_{29} = 2a_{19}$$

$$\text{LHS} \rightarrow a_{29} = a + (29-1)d$$

$$a_{29} = a + 28d$$

$$= -8d + 28d$$

$$a_{29} = 20d$$

$$\text{LHS} = \text{RHS} \quad \text{HP}$$

#LP:

If the seventh term of an A.P. is $\frac{1}{9}$ and its ninth term is $\frac{1}{7}$, find its 63rd term.

[CBSE 2014]

Given,

$$a_7 = \frac{1}{9}$$

$$a_9 = \frac{1}{7}$$

$$a + 6d = \frac{1}{9} \quad \text{--- (I)}$$

$$a + 8d = \frac{1}{7} \quad \text{--- (II)}$$

$$6d - 8d = \frac{1}{9} - \frac{1}{7}$$

$$-2d = \frac{7 - 9}{63}$$

$$+2d = +\frac{2}{63}$$

$$d = \frac{1}{63}$$

$$a + 8d = \frac{1}{7} \quad \text{--- (II)}$$

put in (I)

$$a + 8\left(\frac{1}{63}\right) = \frac{1}{9}$$

$$a + \frac{8}{63} = \frac{1}{9}$$

$$a = \frac{1}{9} - \frac{8}{63}$$

$$a = \frac{7 - 8}{63} = -\frac{1}{63}$$

$$a_{63} = ? ?$$

$$\rightarrow \left(a + 62d \right)$$

$$\Rightarrow \frac{1}{63} + 62\left(\frac{1}{63}\right)$$

$$= \frac{1 + 62}{63} = \frac{63}{63} = 1$$

$$a_{63} = 1$$

#LP: Which term of the A.P. 3, 15, 27, 39, ... will be 120 more than its 21st term?
(Delhi 2019)

AP: 3, 15, 27, 39 - - - $\rightarrow d = 15 - 3 = 12$

Let n^{th} term is 120 more than 21st term

$$\therefore a_n = 120 + a_{21}$$

$$a + (n-1)d = 120 + a + 20d$$

$$(n-1)(12) = 120 + 20(12)$$

$$12n - 12 = 120 + 240$$

$$12n = 360 + 12$$

$$12n = 372 \quad \checkmark$$

$$n = 31$$

$\therefore$ Ans $\rightarrow$ 31st term

~~कोलरी~~ term } let 'n'
~~सिरी~~ term
~~करी~~ term

#LP:

Which term of the sequence -1, 3, 7, 11, ... is 95?

A 20th

B 23rd

☒ C 25th

D 27th

$$\begin{aligned} a_2 - a_1 &= 3 - (-1) = 4 \\ a_3 - a_2 &= 7 - 3 = 4 \\ a_4 - a_3 &= 11 - 7 = 4 \end{aligned} \quad \left. \begin{array}{l} a_1, a_2, a_3, a_4 \\ \text{r.d. same.} \end{array} \right\} \text{Yes } \textcircled{AP} \checkmark$$

let n^{th} term is 95.

$$\text{i.e. } a_n = 95$$

$$a + (n-1)d = 95$$

$$(-1) + (n-1)(4) = 95$$

$$(n-1)4 = 96 \quad 24$$

$$n-1 = 24 \Rightarrow \boxed{n=25} \quad \checkmark \checkmark$$

25th term

#LP:

How many terms are there in the sequence 3, 6, 9, 12, ... 111?

Sequence $\Rightarrow$ 3, 6, 9, 12, ... 111
 a_1, a_2, a_3, a_4

$$a_2 - a_1 = 6 - 3 = 3$$

$$a_3 - a_2 = 9 - 6 = 3$$

$$a_4 - a_3 = 12 - 9 = 3$$

} r.d. are same.
It's AP.

AP: 3, 6, 9, 12, ... 111

let 'n' terms

$$a_n = 111$$

$$a_n = 111$$

$$a + (n-1)d = 111$$

$$3 + (n-1)3 = 111$$

$$(n-1)3 = 111 - 3$$

$$(n-1)3 = 108$$

$$n-1 = 36$$

$$n = 36 + 1$$

$$n = 37$$

$\therefore$ 37 terms are there.

#LP:

Is -150 a term of the A.P. 11, 8, 5, 2, ...?

$$d = 8 - 11$$
$$d = -3$$

Let n^{th} term is -150.

$$a_n = -150$$

$$\Rightarrow a + (n-1)d = -150$$

$$\Rightarrow 11 + (n-1)(-3) = -150$$

$$\rightarrow (n-1)(-3) = -150 - 11$$

$$\rightarrow (n-1)(+3) = +161$$

$$n-1 = \frac{161}{3}$$

$$n = \frac{161}{3} + 1$$

[CBSE 2017]

$$n = \frac{161 + 3}{3}$$

$$n = \frac{164}{3}$$

$$n = 54.666\ldots = \frac{54.66}{\sqrt{164}}$$

Since n should be
a Natural No

$\therefore -150$ is not a term
of this AP.

$$\begin{array}{r} 54.66 \\ 15 \overline{) 164} \\ \underline{15} \\ 14 \\ 12 \\ \underline{12} \\ 20 \\ 20 \\ \underline{20} \\ 4 \end{array}$$

#LP:

The sum of 4th and 8th terms of an A.P. is 24 and the sum of the 6th and 10th terms is 34.
Find the first term and the common difference of the A.P.

$$\begin{array}{c} a_4 + a_8 = 24 \\ \wedge \quad \wedge \\ \underline{a+3d} + \underline{a+7d} = 24 \end{array}$$

$$2a + 10d = 24$$

$$2(a + 5d) = 24$$

$$\boxed{a + 5d = 12} \text{ (I)}$$

$$\begin{array}{c} a_6 + a_{10} = 34 \\ \wedge \quad \wedge \\ \underline{a+5d} + \underline{a+9d} = 34 \end{array}$$

$$2a + 14d = 34$$

$$2(a + 7d) = 34$$

$$\boxed{a + 7d = 17} \text{ (II)}$$

$$\begin{array}{l} a = \\ d = \end{array}$$

$$\begin{array}{l} a = ? \\ d = ? \end{array}$$

#LP: Find the 9th term from the end (towards the first term) of the A.P. 5, 9, 13, ..., 185. (Delhi 2016)

AP: 5, 9, 13, - - - - , 185

Reverse AP:- 185, - - - - - , 13, 9, 5

$$d = 9 - 13$$

$$d = -4$$

$$\begin{aligned} a_9 &= a + 8d \\ &= 185 + 8(-4) \\ &= 185 - 32 \end{aligned}$$

$$a_9 = 153$$

#LP: . How many two digit numbers are divisible by 3? (NCERT, Delhi 2019)

2 digit : 10 ————— 99

div. by 3 $\Rightarrow$ 12, 15, 18, ———— 99

AP

$15 - 12 = 3$
 $18 - 15 = 3$ } Yes

AP: 12, 15, 18, ———— 99 $\leftarrow a_n$

Let 'n' terms

$$\begin{aligned} a_n &= 99 \\ a + (n-1)d &= 99 \\ 12 + (n-1)(3) &= 99 \\ (n-1)(3) &= 99 - 12 \end{aligned}$$

$$(n-1)(3) = 87$$

$$n-1 = 29$$

$$n = 29 + 1$$

$$n = 30$$

$\therefore$ there are 30 two digit numbers div. by 3.

#LP:

. How many three-digit numbers are divisible by 7?

[CBSE 2015, 19]

3 digits no. $\rightarrow$ 100 ————— 999

div by 7 $\rightarrow$ 105, 112, 119, ————, 994

$112 - 105 = 7$
 $119 - 112 = 7$ } const sum. Yes AP

AP: 105, 112, 119, ————, 994

let 'n' term.

$$a_n = 994$$

$$\begin{array}{r} 142 \\ 7 \overline{) 999} \\ \underline{7} \\ 29 \\ \underline{28} \\ 19 \\ \underline{14} \\ 5 \end{array}$$

R $\rightarrow$ 5

$$999 - 5 = 994$$

$$a_n = 994$$

$$a + (n-1)d = 994$$

$$105 + (n-1)(7) = 994$$

$$(n-1)(7) = 994 - 105$$

$$(n-1)(7) = \cancel{889} \quad 127$$

$$n-1 = 127$$

$$n = 127 + 1$$

$$\boxed{n = 128}$$

∴ there are 128 three
digit numbers
div. by 7.

#LP:

Find the number of natural numbers between 101 and 999 which are divisible by both 2 and 5.

[CBSE 2025]

both 2 and 5

$$\boxed{\text{LCM} = 10}$$

101 ————— 999

div by 10 $\Rightarrow$ 110, 120, 130 - - - - 990

$$\begin{array}{l} 120 - 110 = 10 \\ 130 - 120 = 10 \end{array} \left. \begin{array}{l} \text{c.d are same} \\ \text{Yes AP} \end{array} \right\}$$

AP: 110, 120, 130 - - - - 990

let 'n' terms.

$$a_n = 990$$

$\downarrow$ n =

n^{th}

#LP:

Which term of the A.P. 65, 61, 57, 53, ... is its first negative term?

[CBSE 2023]

$$d = 61 - 65$$
$$d = -4$$

let n^{th} term is first -ve term.

$$a_n < 0$$

$$a + (n-1)d < 0$$

$$65 + (n-1)(-4) < 0$$

$$65 - 4(n-1) < 0$$

$$65 < 4(n-1)$$

$$65 < 4n - 4$$

$$65 + 4 < 4n$$

$$69 < 4n$$

$$\frac{69}{4} < n$$

$$\text{or, } n > \frac{69}{4}$$

$$n > 17.25$$

$$n = 18$$

$\therefore 18^{\text{th}}$ term is first -ve term.

$$\begin{array}{r} 17.25 \\ 4 \overline{) 69} \\ \underline{68} \\ 10 \end{array}$$

#LP:

In a given A.P.. If the p th term is q and q th term is p then show that n th term is $p + q - n$

$$a_p = q$$

$$a + (p-1)d = q$$

$$a + pd - d = q \quad \text{--- (I)}$$

$$a + qd - d = p \quad \text{--- (II)}$$

$$pd - qd = q - p$$

$$d(p - q) = q - p$$

$$d(p - q) = -(p - q)$$

$$d = -1$$

$$a_q = p$$

$$a + (q-1)d = p$$

$$a + qd - d = p \quad \text{--- (II)}$$

put in (II)

$$a + q(-1) - (-1) = p$$

$$a - q + 1 = p$$

$$a = p + q - 1$$

$$\text{To show: } a_n = p + q - n$$

$$\text{LHS } a_n = a + (n-1)d$$

$$a_n = (p + q - 1) + (n-1)(-1)$$

$$= p + q - 1 - n + 1$$

$$= p + q - n = \text{RHS}$$

Proved

$$a_{\underline{n}} = a + (\underline{n} - 1)d$$

$$a_{\underline{p}} = a + (\underline{p} - 1)d$$

$$a_{\underline{q}} = a + (\underline{q} - 1)d$$

$$a_{\text{shobhit}} = a + (\text{shobhit} - 1)d$$

#LP:

The mth term of an A.P. is $\frac{1}{n}$ and nth term is $\frac{1}{m}$. Its (mn)th term is :

$$a_m = \frac{1}{n}$$

$$a + (m-1)d = \frac{1}{n}$$

$$a + md - d = \frac{1}{n} \quad \text{--- (I)}$$

$$\begin{array}{r} a + nd - d = \frac{1}{m} \\ - \quad - \quad + \quad - \quad - \\ \hline \end{array}$$

$$md - nd = \frac{1}{n} - \frac{1}{m}$$

$$d(m-n) = \frac{m-n}{nm}$$

$$d = \frac{1}{mn}$$

$$a_n = \frac{1}{m}$$

$$a + (n-1)d = \frac{1}{m}$$

$$a + nd - d = \frac{1}{m} \quad \text{--- (II)}$$

put in (I)

$$a + n\left(\frac{1}{mn}\right) - \left(\frac{1}{mn}\right) = \frac{1}{m}$$

$$a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{m}$$

$$a = \frac{1}{mn}$$

$$a_{mn} = a + (mn-1)d$$

$$a_{mn} = \frac{1}{mn} + (mn-1)\left(\frac{1}{mn}\right)$$

$$= \frac{1}{mn} + \cancel{mn} \times \frac{1}{mn} - 1 \times \frac{1}{mn}$$

$$\cancel{\frac{1}{mn}} + 1 - \cancel{\frac{1}{mn}} = 1$$

Middle Term:



Position of middle term = 3rd position

Value of middle term = $a_3 \Rightarrow a + 2d$

* Position of MT:

if $n \rightarrow \text{odd}$

$$\text{Pos. of MT} = \left(\frac{n+1}{2}\right)^{\text{th}}$$

if $n \rightarrow \text{even}$

$$\text{Pos. of MT} \Rightarrow \left(\frac{n}{2}\right)^{\text{th}} \text{ \& } \left(\frac{n}{2} + 1\right)^{\text{th}}$$

$$n=5$$

$$\frac{\quad}{1^{\text{st}}} \quad \frac{\quad}{2^{\text{nd}}} \quad \frac{a_3}{3^{\text{rd}}} \quad \frac{a_4}{4^{\text{th}}} \quad \frac{\quad}{5^{\text{th}}} \quad \frac{\quad}{6^{\text{th}}}$$

↓ ↓
middle terms

$$\begin{aligned} \text{Pos} &= \left(\frac{n}{2}\right)^{\text{th}} \& \left(\frac{n}{2} + 1\right)^{\text{th}} \\ &= \left(\frac{5}{2}\right)^{\text{th}} \& \left(\frac{5}{2} + 1\right)^{\text{th}} \\ &\rightarrow \underline{\underline{3^{\text{rd}} \& 4^{\text{th}}}} \end{aligned}$$

Value of m

$$a_3 = a + 2d$$

$$a_4 = a + 3d$$

#LP: Find the middle term of the A.P. 6, 13, 20, ..., 216. (Delhi 2015)

- ① $n = ?$
- ② position of MT
- ③ value of MT

AP: 6, 13, 20, ..., 216 $\leftarrow n^{\text{th}}$

$$\begin{aligned} d &= 13 - 6 \\ d &= 7 \end{aligned}$$

let 'n' terms

$$a_n = 216$$

$$a + (n-1)d = 216$$

$$6 + (n-1)7 = 216$$

$$(n-1)7 = 216 - 6$$

$$(n-1) \times 7 = 210 \quad 30$$

$$n-1 = 30$$

$$n = 31$$

— — — — — 31 term.

$$n = 31$$

odd

Position of MT

$$MT = \left(\frac{n+1}{2} \right)^{\text{th}}$$

$$= \left(\frac{31+1}{2} \right)^{\text{th}} = \left(\frac{32}{2} \right)^{\text{th}} = 16^{\text{th}}$$

AP:



31 terms

$$a_{16} = a + 15d$$
$$= 6 + 15(7)$$

$$= 6 + 105$$

$$a_{16} = 111$$

Value of
middle term

AP

① n^{th} term of AP

$$a_n = a + (n-1)d$$

② Sum of first n terms of AP.

Sum of first ~~n~~ terms of AP:

AP: →
n terms sum

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} [a + \underbrace{a + (n-1)d}_{q_n}]$$

$$= \frac{n}{2} (a + a_n)$$

when last
term is
given

$$S_n = \frac{n}{2} (a + l)$$

l = last term.

LP: AP: 2, 4, 6, 8 - - -

Find sum of first 11 terms of AP!

$$S_{11} = \frac{11}{2} (2a + (11-1)d)$$

$$= \frac{11}{2} [2(2) + 10(2)]$$

$$= \frac{11}{2} [4 + 20]$$

$$= \frac{11}{2} \times 24 = 132$$

A diagram showing the sum of an arithmetic series S_n branching into two equivalent formulas. The term S_n is enclosed in a circle on the left. A vertical line extends from the circle, with two arrows pointing to the right. The top arrow points to the formula $\frac{n}{2} [2a + (n-1)d]$, and the bottom arrow points to the formula $\frac{n}{2} [a + l]$.

$$S_n \rightarrow \frac{n}{2} [2a + (n-1)d]$$
$$S_n \rightarrow \frac{n}{2} [a + l]$$

#LP: Find the sum of the first 16 terms of an Arithmetic Progression whose 4th and 9th terms are -15 and -30 respectively. (2020)

$$\begin{aligned} a_4 = -15 &\Rightarrow d + 3d = -15 \quad \text{--- (I)} \\ a_9 = -30 &\Rightarrow a + 8d = -30 \quad \text{--- (II)} \\ \hline &\quad - \quad + \\ &\quad - \cancel{8}d = + \cancel{15} \quad 3 \\ &\quad \boxed{d = -3} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{put in (I)} \quad a + 3(-3) &= -15 \\ a - 9 &= -15 \\ a &= 9 - 15 \\ \boxed{a} &= \boxed{-6} \quad \checkmark \end{aligned}$$

$$\begin{aligned} S_{16} &= \frac{16}{2} (2a + (16-1)d) \\ S_{16} &= 8 (2(-6) + 15(-3)) \\ &= 8 (-12 - 45) \\ &= 8 \times -57 \\ \boxed{S_{16}} &= \boxed{-456} \end{aligned}$$

#LP: Find the sum of the first 100 natural numbers. (2020)

$$N: 1 + 2 + 3 + 4 - - -$$

$n = 100$

$$S_{100} = \frac{100}{2} (2(1) + (100-1)(1))$$

$$= \frac{100}{2} (2 + 99)$$

$$= \frac{50 \times 101}{2} = \underline{\underline{5050}}$$

#LP:

Find the sum of first n natural numbers.

$$1 + 2 + 3 + 4 + \dots$$

$n = n$

$$S_n = \frac{n}{2} (2(1) + (n-1)(1))$$

$$= \frac{n}{2} (2 + n - 1)$$

$$S_n = \frac{n}{2} (n+1)$$

$$S_n = \frac{n(n+1)}{2}$$

#LP: If the sum of first 7 terms of an A.P. is 49 and that of its first 17 terms is 289, find the sum of first n terms of the A.P. (NCERT, Delhi 2016)

$$S_7 = 49$$

$$\cancel{7} \frac{7}{2} [2a + (7-1)d] = \cancel{49} 7$$

$$2a + 6d = 7 \times 2$$

$$\cancel{2} (a + 3d) = \cancel{14} 7$$

$$\boxed{a + 3d = 7} \text{--- (I)}$$

$$S_{17} = 289$$

$$\cancel{17} \frac{17}{2} [2a + (17-1)d] = \cancel{289} 17$$

$$2a + 16d = 17 \times 2$$

$$\cancel{2} (a + 8d) = 17 \times \cancel{2}$$

$$\boxed{a + 8d = 17} \text{--- (II)}$$

$$\left. \begin{array}{l} a = \\ d = \end{array} \right\}$$

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

put

#LP: Solve: $1+4+7+10+\dots+x=287$ (2020)

$$\left. \begin{array}{l} 4-1 \Rightarrow 3 \\ 7-4 \Rightarrow 3 \\ 10-7 \Rightarrow 3 \end{array} \right\} \text{all c.d are same} \\ \therefore \text{series is } \textcircled{\text{AP}}$$

$\textcircled{\text{AP}}$ $1+4+7+10+\dots+x=287$

let 'n' terms

$$\begin{aligned} a_n &= x \\ a+(n-1)d &= x \\ 1+(n-1)(3) &= x \\ (n-1)(3) &= x-1 \\ n-1 &= \frac{x-1}{3} \end{aligned}$$

$$\begin{aligned} n &= \frac{x-1}{3} + 1 \\ n &= \frac{x-1+3}{3} \\ n &= \frac{x+2}{3} \end{aligned}$$

$$? \quad \frac{n}{2}(a+l) = 287$$

$$\Rightarrow \frac{\left(\frac{x+2}{3}\right)(1+x)}{2} = 287$$

$$\Rightarrow \frac{(x+2)(x+1)}{6} = 287$$

$$\times (x+2)(x+1) = 287 \times 6$$

$$\Rightarrow x^2 + x + 2x + 2 = 1722$$

$$\times x^2 + 3x = 1722 - 2$$

$$\boxed{x^2 + 3x - 1720 = 0}$$

$$x^2 + 3x - 1720 = 0$$

$$x^2 - 40x + 43x - 1720 = 0$$

$$x(x-40) + 43(x-40) = 0$$

$$(x-40)(x+43) = 0$$

$$x-40=0$$

$$x=40$$

$$n = \frac{x+2}{3} = \frac{40+2}{3} = \frac{42}{3} = 14$$

$$x+43=0$$

$$x = -43$$

$$n = \frac{x+2}{3} = \frac{-43+2}{3} = -\frac{41}{3}$$

$$p+q = 3$$

$$pq = -1720$$

2	1720
2	860
2	430
5	215
43	43
	1

#LP:

Find the sum:

$$18 + 15\frac{1}{2} + 13 + \dots + (-49\frac{1}{2})$$

check AP

$$18 + \frac{31}{2} + 13 + \dots + \left(-\frac{97}{2}\right)$$

2 + 'n' term

$$a_n = -\frac{97}{2}$$

n = "

$$S_n = \frac{n}{2}(a + l)$$

#LP: Find the sum of n terms of the series

$$\left(4 - \frac{1}{n}\right) + \left(4 - \frac{2}{n}\right) + \left(4 - \frac{3}{n}\right) + \dots - n \text{ terms} \quad (\text{Delhi 2017})$$

$$(4 + 4 + 4 + \dots - n \text{ times}) - \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \frac{4}{n} + \dots - n \text{ times}\right)$$

$$\Rightarrow 4n - \left(\frac{1+2+3+\dots+n}{n}\right) \xrightarrow{\text{'n' natural no. sum}} \Rightarrow 4n - \left(\frac{n+1}{2}\right)$$

$$\Rightarrow 4n - \left(\frac{\cancel{n}(n+1)}{2}\right) \Rightarrow \frac{8n - n - 1}{2} \rightarrow \frac{7n-1}{2}$$

#LP: Show that the sum of all terms of an A.P. whose first term is a, the second term is b and the last term is c is equal to $\frac{(a+c)(b+c-2a)}{2(b-a)}$. (CBSE 2020)

(AP): $a, b, \dots, c \leftarrow n^{\text{th}}$

$c.d = (b-a)$

let 'n' term.

$$a_n = c$$

$$A + (n-1)D = c$$

$$a + (n-1)(b-a) = c$$

$$(n-1)(b-a) = c-a$$

$$n-1 = \frac{c-a}{b-a}$$

$$n = \frac{c-a}{b-a} + 1$$

$$n = \frac{c-a + b-a}{b-a}$$

$$n = \frac{b+c-2a}{b-a}$$

$$S_n = ? \frac{n}{2} (a+l)$$

$$S_n = \frac{(b+c-2a)(a+c)}{(b-a) \times 2}$$

$$S_n = \frac{(b+c-2a)(a+c)}{2(b-a)}$$

HP

#Q:- If n^{th} term of AP is given by $3n^2 - 4$, then find
it's 6^{th} term & sum of first 8 terms.

given $\rightarrow a_n = 3n^2 - 4$

$n=1 \rightarrow a_1 = 3(1)^2 - 4 \Rightarrow -1 = a$

$n=2 \rightarrow a_2 = 3(2)^2 - 4 \Rightarrow 3(4) - 4$
 $\Rightarrow 8$

$\therefore a = -1$

$d = a_2 - a_1 \Rightarrow 8 - (-1)$

$d = 9$

$a_6 = a + 5d$

$= -1 + 5(9) \Rightarrow -1 + 45 \Rightarrow 44$

$S_8 = \frac{8}{2} (2a + (8-1)d)$

$= 4 (2(-1) + 7(9))$

$= 4 (-2 + 63)$

$= 4 \times 61 \Rightarrow 244 = S_8$

#LP: If the sum of first m terms of an AP is $2m^2 + 3m$, then its second term is :

(1 Mark) (2025)

(A) 10

(B) 9

(C) 12

(D) 4

$$S_m = 2m^2 + 3m$$

$$m=1 \rightarrow S_1 = 2(1)^2 + 3(1) = 5 = a_1$$

$$m=2 \rightarrow S_2 = 2(2)^2 + 3(2) \\ = 2 \times 4 + 6 \\ = 14$$

$$S_2 = 14$$

$$a_1 + a_2 = 14$$

$$5 + a_2 = 14$$

$$a_2 = 14 - 5$$

$$a_2 = 9$$

$$d = a_2 - a_1$$

$$= 9 - 5$$

$$d = 4$$

#LP:

If the sum of the first n terms of an A.P. is $\frac{1}{2}(3n^2 + 7n)$,
then find its n^{th} term. Hence write its 20th term.
(Delhi 2015)

$$S_n = \frac{1}{2}(3n^2 + 7n)$$

$$n=1 \rightarrow S_1 = \frac{1}{2}(3(1)^2 + 7(1)) = \frac{1}{2} \times 10 \rightarrow \boxed{5 = a_1} = a \quad \checkmark$$

$$n=2 \rightarrow S_2 = \frac{1}{2}(3(2)^2 + 7(2)) = \frac{1}{2} \times (12 + 14) \\ = \frac{1}{2} \times 26$$

$$S_2 = 13$$

$$a_1 + a_2 = 13$$

$$5 + a_2 = 13 \rightarrow a_2 = 13 - 5 \quad \checkmark$$

$$\boxed{a_2 = 8}$$

$$d = a_2 - a_1 \\ = 8 - 5$$

$$\boxed{d = 3}$$

$$a_n = a + (n-1)d$$

$$a_n = 5 + (n-1)(3)$$

$$a_n = 5 + 3n - 3$$

$$\boxed{a_n = 3n + 2} \quad \checkmark$$

$$a_{20} = 3(20) + 2 \\ = 60 + 2$$

$$\boxed{a_{20} = 62} \quad \checkmark$$

#LP: How many terms of the A.P. 18, 16, 14, ... be taken so that their sum is zero?
(Delhi 2016)

AP: $18 + 16 + 14 + \dots = 0$

$$d = 16 - 18$$

$$d = -2$$

Let 'n' terms

$$S_n = 0$$

$$\rightarrow \frac{n}{2} [2(18) + (n-1)(-2)] = 0$$

$$\rightarrow \frac{n}{2} [36 - 2n + 2] = 0$$

$$\rightarrow \frac{n}{2} [38 - 2n] = 0$$

$$\frac{n}{2} = 0$$
$$n = 0$$

$$38 - 2n = 0$$

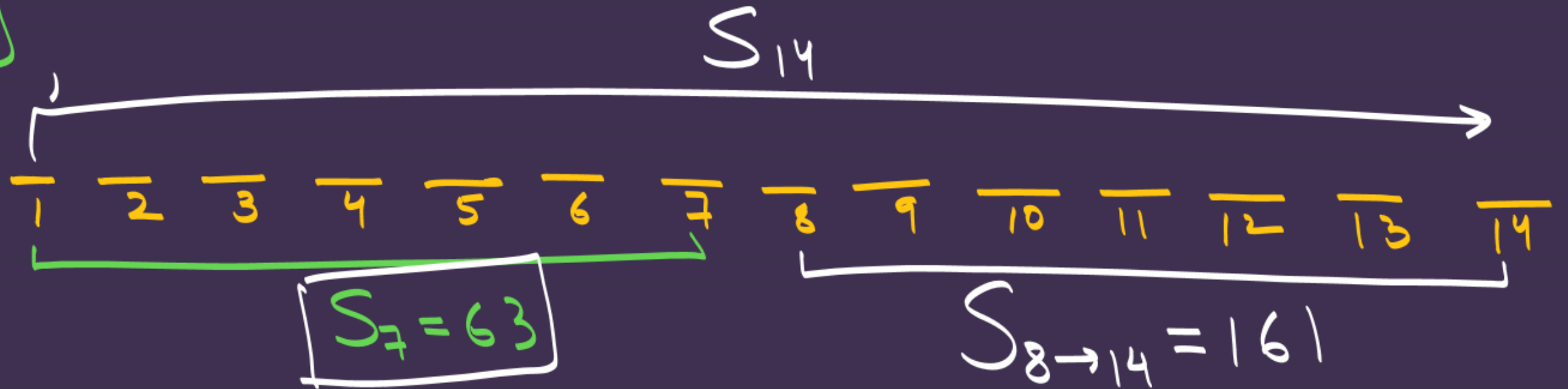
$$19 \cdot \cancel{2} = \cancel{2} n$$

$$\boxed{n = 19}$$

#LP:

The sum of the first 7 terms of an A.P. is 63 and the sum of its next 7 terms is 161. Find the 28th term of this A.P. [CBSE 2014]

$$S_7 = 63$$



$$S_{14} - S_7 = 161$$

$$S_{14} - 63 = 161$$

$$S_{14} = 161 + 63$$

$$S_{14} = 224$$

a_{28}
 $a + 27d$
?

$$S_7 = 63$$

$$\cancel{\frac{7}{2}} (2a + (7-1)d) = \cancel{63}^9$$

$$2a + 6d = 9 \times 2$$

$$\cancel{2}(a + 3d) = 9 \times \cancel{2}$$

$$\boxed{a + 3d = 9} \text{ (I)}$$

$$S_{14} = 224$$

$$\cancel{\frac{14}{2}} (2a + (14-1)d) = \cancel{224}^{112}$$

$$2a + 13d = \cancel{224}^{\cancel{7}} 32$$

$$\boxed{2a + 13d = 32} \text{ (II)}$$

$$\left. \begin{array}{l} a = \\ d = \end{array} \right\}$$

$$a_{28} = a + 27d$$

put

#LP: The p th, q th and r th terms of an A.P. are a , b and c respectively. Show that $a(q - r) + b(r - p) + c(p - q) = 0$.

$$a_p = a$$

$$A + (p-1)D = a$$

$$\boxed{A + pD - D = a}$$

$$a_q = b$$

$$A + (q-1)D = b$$

$$\boxed{A + qD - D = b}$$

$$a_r = c$$

$$A + (r-1)D = c$$

$$\boxed{A + rD - D = c}$$

(LHS)

$$a(q-r) + b(r-p) + c(p-q)$$

$$\Rightarrow \frac{(A+pD-D)(q-r)}{1} + \frac{(A+qD-D)(r-p)}{1} + \frac{(A+rD-D)(p-q)}{1}$$

$$\Rightarrow \cancel{A} + pD - D - \cancel{A} - \cancel{pD} + D - \cancel{A} - \cancel{qD} + D - \cancel{A} - \cancel{rD} + D - \cancel{A} - \cancel{pD} + D + \cancel{A} + \cancel{qD} - \cancel{rD} - \cancel{A} - \cancel{pD} + D + \cancel{A} + \cancel{rD} - \cancel{pD} - \cancel{A} - \cancel{qD} + D + \cancel{A} + \cancel{pD} - \cancel{qD} - \cancel{A} - \cancel{rD} + D$$

$$\Rightarrow 0 = \text{RHS (HP)}$$

#LP: The sums of first n terms of three arithmetic progressions are S_1 , S_2 and S_3 respectively. The first term of each A.P. is 1 and their common differences are 1, 2 and 3 respectively. Prove that $S_1 + S_3 = 2S_2$ (AI 2016)

$S_1, d_1 = 1$

$S_1 = \frac{n}{2} (2(1) + (n-1)(1))$

$S_1 = \frac{n}{2} (2 + n - 1)$

$S_1 = \frac{n}{2} (n+1)$ ✓

$S_2, d_2 = 2$

$S_2 = \frac{n}{2} (2(1) + (n-1)(2))$

$S_2 = \frac{n}{2} (2 + 2n - 2)$

$S_2 = \frac{n}{2} (2n)$

$S_2 = n^2$ ✓

$S_3, d_3 = 3$

$S_3 = \frac{n}{2} (2(1) + (n-1)(3))$

$S_3 = \frac{n}{2} (2 + 3n - 3)$

$S_3 = \frac{n}{2} (3n - 1)$ ✓

To prove: $S_1 + S_3 = 2S_2$

LHS

$$\rightarrow \frac{n}{2} \underbrace{(n+1)} + \frac{n}{2} \underbrace{(3n-1)}$$

$$\rightarrow \frac{n}{2} [n+1+3n-1]$$

$$\rightarrow \frac{n}{2} [4n]$$

$$\boxed{\text{LHS} = 2n^2}$$

$$\text{RHS} = 2S_2$$

$$\boxed{\text{RHS} = 2n^2}$$

HP

#LP: The sum of the third term and the seventh term of an AP is 6 and their product is 8. Find the sum of the first sixteen terms of the AP. [CBSE 2025]

$$\begin{aligned}a_3 + a_7 &= 6 \\ \downarrow \quad \downarrow \\ \underline{a+2d} + \underline{a+6d} &= 6 \\ 2a + 8d &= 6 \\ \cancel{2}(a + 4d) &= \cancel{6}^3 \\ \boxed{a + 4d = 3} \quad \text{--- (I)} \\ \rightarrow \underline{a = 3 - 4d}\end{aligned}$$

$$\begin{aligned}a_3 \cdot a_7 &= 8 \\ \hat{a+2d} \hat{a+6d} &= 8 \\ \Rightarrow a^2 + 6ad + 2ad + 12d^2 &= 8 \\ \Rightarrow \underline{a^2} + 12d^2 + 8\underline{ad} &= 8 \\ \Rightarrow (3-4d)^2 + 12d^2 + 8(3-4d)(d) &= 8 \\ \Rightarrow \underline{9} + \underline{16d^2} - \cancel{24d} + \underline{12d^2} + \cancel{24d} - \underline{32d^2} &= 8 \\ \Rightarrow 28d^2 - 32d^2 + 9 - 8 &= 0 \\ \Rightarrow -4d^2 + 1 &= 0 \\ \Rightarrow +4d^2 &= +1 \quad \rightarrow d^2 = \frac{1}{4} \\ &\quad \boxed{d = \pm \frac{1}{2}}\end{aligned}$$

$$S_{16} = \frac{16}{2} (2\underline{a} + (16-1)\underline{d})$$

✓

$$d = \pm \frac{1}{2}$$

$$d = +\frac{1}{2}$$

$$a = 3 - 4\left(\frac{1}{2}\right)$$

$$a = 1$$

$$\therefore S_{16} = \frac{16}{2} \left(2(1) + (16-1)\left(\frac{1}{2}\right) \right)$$



$$d = -\frac{1}{2}$$

$$a = 3 - 4\left(-\frac{1}{2}\right)$$

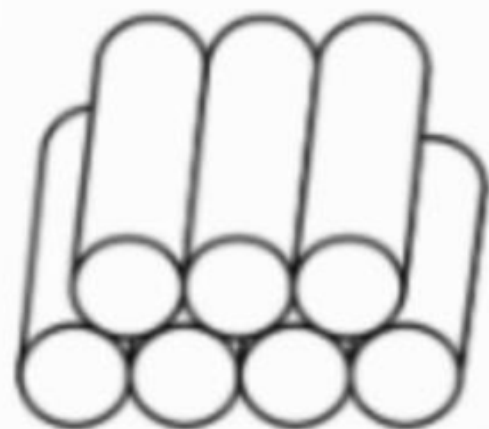
$$a = 3 + 2 = 5$$

$$S_{16} = \frac{16}{2} \left(2(5) + (16-1)\left(-\frac{1}{2}\right) \right)$$



#LP:

22 logs in the bottom row, 21 in the next row, 20 in the row next to it and so on (as shown by an example). In how many rows, are the 250 logs placed and how many logs are there in the top row? (2023)



→ 19 ← $n=4$
→ 20 ← $n=3$
→ 21 ← $n=2$
→ 22 ← $n=1$

Let 'n' rows

$$S_n = 250$$

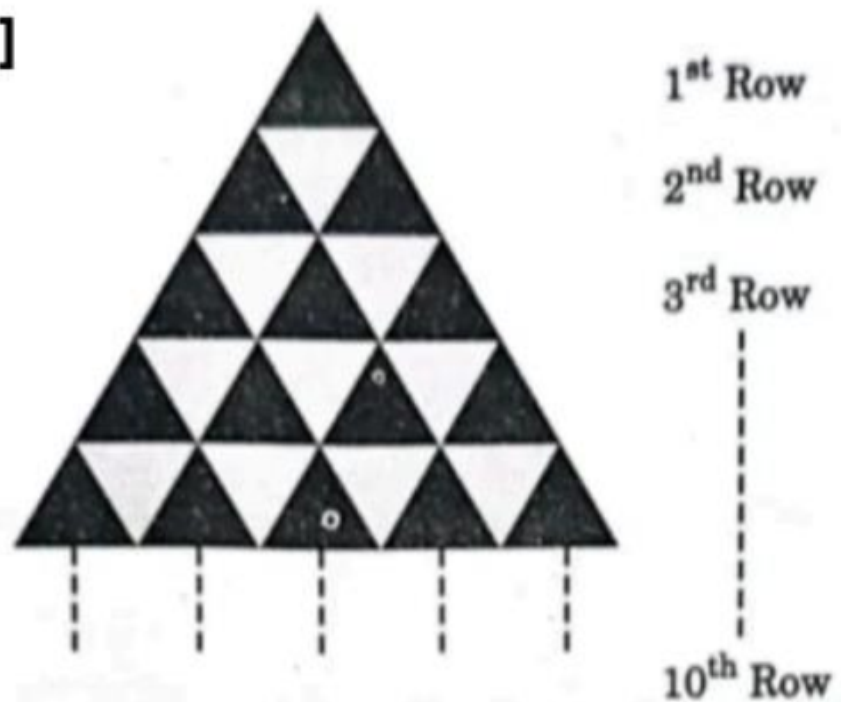
$$\frac{n}{2} (2(22) + (n-1)(-1)) = 250$$

↓
 $n=?$ ✓

$$(ii) a_n = a + (n-1)d$$

37. In an equilateral triangle of side 10 cm, equilateral triangles of side 1 cm are formed as shown in the figure below, such that there is one triangle in the first row, three triangles in the second row, five triangles in the third row and so on.

[CBSE 2025]



Based on given information, answer the following questions using Arithmetic Progression.

- How many triangles will be there in bottom most row?
- How many triangles will be there in fourth row from the bottom?
- (a) Find the total number of triangles of side 1 cm each till 8th row.

OR

- (b) How many more number of triangles are there from 5th row to 10th row than in first 4 rows? Show working.

$$\text{No. of } \Delta s \rightarrow 1, 3, 5, \dots, 19$$

789 10th row

$$\begin{aligned} \text{(i)} \quad a_{10} &= a + 9d \\ &= 1 + 9(2) = 19 \text{ triangles} \end{aligned}$$

$$\text{(ii)} \quad a_7 = a + 6d$$

$$\text{(iii)} \quad S_8 = \frac{8}{2} (2(1) + (8-1)(2))$$

$$\text{(b)} \quad S_{5 \rightarrow 10} = S_{10} - S_4$$



$$1b) \frac{(S_{10} - S_4) - S_4}{(S \rightarrow 10) - (\text{first } 4)}$$

$$\Rightarrow S_{10} - 2S_4$$

$$\approx \frac{10}{2} [2(1) + (10-1)(2)] - 2 \left[\frac{4}{2} (2(1) + (4-1)(2)) \right]$$

14/14 Completed!
Thank you Goodies !!



15 Feb]
16 Feb]

Maha Marathon

4 pm